

Haocheng He

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PROFESSIONAL SUMMARY

Aviation Engineering MSc candidate at PolyU (GPA: 3.95/4.3), with an Industrial Engineering undergraduate and a Law minor from GDUT. Knowledge spans engineering optimization, airport simulation, AI hardware, and intellectual property. Applied lean methods to improve line balance from 58% to 70%; built AnyLogic airport simulation models comparing 5 routing algorithms across 4 emergency scenarios (D* Lite reliability index: 1.342); contributed to Ascend 310B smart hardware R&D. 3 published invention patent applications.

EDUCATION

The Hong Kong Polytechnic University <i>Master of Science in Aviation Engineering and Operations Management</i> GPA: 3.95 / 4.3 (Current) Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty	Hong Kong SAR, China Sep 2025 – Jan 2027
Guangdong University of Technology <i>Bachelor of Engineering in Industrial Engineering</i> GPA: 3.67 (Top 20%) Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports Honors: University-level Excellent Graduation Thesis, Graduation Design Innovation Award	Guangzhou, China Sep 2021 – Jun 2025
<i>Bachelor of Laws (Minor Degree in Law)</i> GPA: 3.61 Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management	Sep 2022 – Jun 2025

EXPERIENCE

Assistant AI Engineer <i>Shenzhen Wuxian Hongyin Technology Co., Ltd.</i> - Participated in Ascend 310B smart piano R&D across 3 development phases; tested 10+ boards (CPLD, STM32), iterated motor driver 3 times, developed firmware and authored 5 technical documents. - Participated in camera- and Ascend-based piano fingering recognition module integration (Python/C++); diagnosed cross-layer issues. - Maintained Ubuntu servers, configured Huawei network equipment, operated 3D printer, supported technical interviews.	May 2025 – Jun 2026 Shenzhen, China
Research Assistant <i>Guangdong University of Technology (with Guangzhou Baiyun Int’l Airport)</i> - Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios. - Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms. - Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).	Sep 2023 – Jun 2025 Guangzhou, China
Industrial Engineering Intern <i>Midea Group — Kitchen & Water Heater Division</i> - Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps. - Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.	Apr 2024 – May 2024 Foshan, China
Manufacturing Operations Intern <i>Guangxi LiuGong Machinery Co., Ltd.</i> - Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis. - Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation.	Jul 2023 – Aug 2023 Liuzhou, China

PROJECTS

Aircraft Engine Remaining Useful Life Prediction (Course Project) <i>Guangdong University of Technology</i> - Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA. - Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.	Apr 2024 – Jul 2024 Guangzhou
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SKILLS & INTELLECTUAL PROPERTY

Programming: Python, Java, SQL/MySQL, C (Embedded/STM32), JS/TS Workflows, Prompt Eng.	ML: LSTM, BiLSTM, Transformer, RF, PCA	AI: LLM		
Simulation: AnyLogic, Plant Sim., OR-Tools, Dijkstra, A*, BFS, GA, ECRS	Tools: Ubuntu/Linux, Git, JLC EDA, Oscilloscope, Logic Analyzer			
Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)				
Patent Applications: <i>MCMC Sampling-Based Motion Control Method for a PI Coating Machine Controller</i> (CN122085794A, 2026); <i>Coating Control Method and System Based on Deep Reinforcement Learning</i> (CN121325552A, 2025); <i>Energy Regulation Method and System for a Household Solar Energy Storage System</i> (CN117767437A, 2024).				
Utility Model: <i>Integrated Equipment for Wire Harness Production</i> (CN221352455U, 2024). SW: Financial Data Analysis (2024SR0146827).				